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Agri Assist: An AI Integrated Farmer Assistant

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Abstract

In this paper, an AI Agriculture Assistant integrating a crop recommendation system and an agricultural query chatbot is developed to provide guidance on crop cultivation. Based on this, the crop recommendation system uses a stack ensemble model comprising Random Forest and Gradient Boosting, and an accuracy of 99.32% and F1-Score of 99.26%. Here, word embeddings are performed through FastText which allows for a quick response time (0.0244 seconds) and a cosine similarity of 0.88629 for the chatbot. With RSA encryption for securing user data, we ensure secure communication. It uses the critical elements of the soil composition, weather and crop performance in formulating tailored agricultural guidance. Back end communication is done through Flask, and the entire system was deployed in this manner with a web interface to drive usage and for real time data responses. This entire application helps the farmers in decision making regarding crops cultivation and management through accurate and secure assistance.

Keywords: Agriculture; AI Assistant; Crop Recommendation; Chatbot; FastText

1. Introduction

Agriculture has been long the backbone of economies and in developing countries especially, it has contributed greatly to food security, employment and GDP. As global populations grow and traditional farming practices are infiltrated by climate change, it is critically important we have solutions that rely on advanced, data-driven technologies to help optimize crop yield, lower costs and foster sustainable agricultural operations. While the most immediate application of technological advances — particularly in artificial intelligence (AI) and machine learning (ML) — to agriculture may be the provision of intelligent decision support systems that aid farmers, there may be longer term benefits as well.

AI helps comprehensively in agriculture in several ways, including in checking the diseases and pests that affects plants, checking the maturity level of crops, the quality of soil and the weather condition, help in making utilization

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of resources and in getting the best returns on investment [1]. They also identify various issues concerning supply and certain strategies of staffing as well as the anticipation of market demands.

This is especially important in determining where it is suitable to cultivate crops and the time of the year, application of fertilizer and pesticides, among other matters that would help in estimating productions and therefore cutting down on expenses like water and fertilizer. The hitches such as weather are being avoided through this, it helps markets and it plays a big role in planning for the farmers' budgets [2].

A chatbot is the type of software program that can actually replicate the human communication via using the text or speech communication [3]. It makes use of the artificial intelligence (AI) for understanding and responding to user inputs often in an attempt to provide a service, respond to questions or ease the process. Several applications, including messaging apps, websites and customer support portals can incorporate the chatbots [4].

chatbots are of significance to farming organizations if they need to step up production or organization and also in decision making. Some of the advantage are; Access to information on crop management, pests, climate and price of the produce in the market, this assist in economical and effective use of resources as well as supervision. It is critical to understand that chatbots simplify each day's tasks, introduce new technologies and market information, and help increase the level of readiness to respond in case of an emergency and allocate resources [5].

As chatbots become integral in handling sensitive user data and performing critical tasks, robust security measures are essential to protect against cyber threats and ensure user trust [6]. Cryptographic techniques play a important role in security framework by providing mechanisms for data confidentiality, integrity, and authenticity. [7]. Encryption (both symmetric and asymmetric) ensures that data remains private and accessible only to authorized parties, while hash functions and digital signatures verify data integrity and authenticate identities, preventing tampering and impersonation. Secure communication protocols further enhance protection by encrypting data in transit, making cryptographic algorithms indispensable for maintaining the security and reliability of chatbot interactions.

This study will work on the following aspects:

- Implementation of a stacked ensemble model for crop recommendation.
- Utilisation of NLP techniques and cryptography algorithm to develop the chabot with data encryption.
- Creation of an application with integrated solution that adds these technologies to offer the intelligent, safe, and convenient help to the farmers to improve the decision making.

The development of this AI Agriculture Assistant includes the development of a crop recommendation system as an assistant to a Chatbot for agricultural queries is presented in this paper. Based on the crop recommendations it employs a stacked ensemble model to increase predictive accuracy and the use of FastText for rapid and efficient query resolution. RSA encryption is used throughout for secure communication and user data privacy on the system. The proposed assistant combines highly effective machine learning techniques and robust security measures to provide predictable, personalized agricultural guidance that may improve the farmers' decision making.

This study is divided into multiple sections. In Section I, an overview of the proposed work is presented, Section II discusses the literature review, Section III describes the data used in this study, Section IV describes the proposed methodology, Section V provides discussion on results of the proposed method and comparison with models, and Section VI presents the conclusion and future work.

2. LITERATURE REVIEW

In recent years, agricultural chatbots and crop recommendation systems have emerged as pivotal tools in enhancing farming efficiency and decision-making processes. These systems leverage machine learning and AI techniques to provide personalized advice and recommendations tailored to farmers' specific needs and regional conditions.This literature review aims to explore different approaches, advancements, and challenges in collaborative filtering, seeking to summarize and evaluate the existing research in this area. Rao et al [8] conducted a comparative study of supervised learning techniques such as KNN, Random Forest and Decision Tree where they used a data set of 22 varieties of crops regarding Random forest present higher accuracy. This paper contributes by providing insights into the effectiveness of machine learning techniques for crop prediction, emphasizing the importance of precision, recall, and F1 Score in model evaluation. Based on Gini and Entropy criteria, the paper provide the analysis of each model presented helps in choosing the most accurate one out of the numerous models used in crop recommendation. Shah et al [9] conducts a comparative analysis of different supervised machine learning prediction, including voting regression, K-Nearest Neighbour (KNN), Support Vector Machine (SVM), Linear Descriptive Analysis (LDA), Least Absolute Shrinkage & Selection Operator (LASSO) regression, and Nearest Centroid Classifier, to forecast appropriate crops, with KNN and SVM achieving the highest accuracy of 97.78%.The research work therefore seeks to identify the robust machine learning algorithm that will be more effective in suggesting right crop to grow given certain sets of data.

The paper highlights the need for a chatbot in the education domain to provide visitor satisfaction.Ranoliya et al [10] presents a design of a chatbot that efficiently answers queries based on FAQs using AIML, that ensures user satisfaction. The chatbot is designed for universities and intended to engage with curious stuternd with several academic related questions. Hiremath et al [11] suggested about an automated system for the education sector, which aims to deliver answers to the information requests of the users through machine learning, natural language processing, pattern matching and data processing algorithms. In this study, there is the introduction of the chatbots which may include chatter-robots that function in areas that involve auto replies or functioning through knowledge which might be input in advance.

Niranjan et al [12] aims to develop an Agriculture Question Answer System to obtain accurate answers for a farmer query through use of machine learning classification. The solution includes training of the system, with the help of RNN sequence to sequence algorithm, the generation of the dataset in the format of XML, stressing the effectiveness of the RNN option as compared to other machine learning tools, such as SVMS and logistic regression. Momaya et al [13] described a farming help chatbot named “Krushi -The Farmer Chatbot” that seeks to foster an improvement in decision support in farming with an aim of boosting crop yield through answering typical farmer’s questions on subjects like crop sowing, diseases or seasonality. The chatbot learning model is an end-to-end trainable model based on AI and ML and the dataset obtained from Kisan Call Centre (KCC) on answering basic and specific queries regarding weather conditions, protection of plants & animals, market prices, use of fertilizers, government schemes, and soil testing with an average accuracy of about 96.1% using RASA X.

Suebsombut et al [14] designed a LINE chatbot application that helps farmers in a decision making process concerning the cultivation of crops including irrigation, fertilizer, disease, insect pests, and weeds control. The study therefore merges the chatbot with a recommendation system and IoT technologies with the purpose of offering the farmer information, knowledge as well as recommendations concerning crop growing. Dhivvya et al [15] demonstrate competence in several tasks aimed at learning English, for instance, question and dialogue generation or the attainment of high Bi-lingual Evaluation Understudy (BLEU) benchmarks hence exemplifying its efficacy in different learning environments. The study made use of datasets with word samples, dialogues for model fine-tuning as well as messenger exchanges that actually occurred in real-life in order to teach chatbots how to comprehend emotions. This underlines how significant it is.

Kuriakose et al [16] uses Deep Learning techniques especially Long Short-Term Memory (LSTM) to develop an analytical model for determining the crop yield based on the type of soil, climate and rainfall to support the farmers to maximize yields on their farms. This study compares the performance of Deep Learning models as compared to the regression models, in the context of crop yield prediction and remarks on how Deep Learning outperforms the regression models. Damaraju et al [17] suggested the image based classification as it has given the best results rather visual classifications in plant disease detection. This study used Hyperspectral imaging for plant disease detection through capturing Spectral signatures that are unique to diseased regions. The study proposed different that have been recommended for use in disease diagnosis in plants, including deep convolutional neural networks (CNNs) where diseases like powdery mildew on grape leaves were diagnosed with high accuracy.

Sonawane et al. [18] have outlined a methodology that have essential recourse in deep learning especially using LSTM models the project seeks to design a personalized chatbot that may be used in the student portal platforms for generating autonomous prompts. The study enhances the generation of text depending on LSTM models thereby making experience to be even better in an application. The study has improved the accuracy to 92%, with the precision and flexibility of the results underlined by the fact that when the measures have been presented in different formats, the accuracy has still ranged between 91 and 93%. During training of the chatbots, 86% of interaction rates were achieved.

Malaka et al [19] provides insight how to use chatbots in a secure environment and sheds light on how their design, implementation, and usage practices can affect metadata security.The paper shows how securely built chatbots can bolster security and enhance user experience by securing critical metadata. In early identification of diabetes in females Oliullah et al [20] has proposed a wide range of machine learning techniques such as random forest classifiers, XGBoost, NGBoost, Bagging, LightGBM, AdaBoost classifiers. The strength of the study is that it predicted diabetes with 92.91% accuracy rate, which is competitive with current methods.

Kalabarige et al [21] presented a multilayer stacked ensemble model to identify and stop phishing websites.Succeeded baseline models in terms of F1-score measurements and accuracy, achieving noteworthy performance with an accuracy range from 96.79% to 98.90% when examined with diverse datasets. Shah et al [22] presented a Privacy Protected Modified Double Ratchet Algorithm for Secure Chatbot Application. The possible risks and weaknesses in the current chatbot designs are examined in this research.

The existing research has predominantly focused on evaluating and implementing machine learning algorithms such as KNN, Decision Trees, and Random Forest for crop recommendation systems. Our research aims to fill this gap by exploring and implementing stacked ensemble model to improve the performance of crop recommendation systems, thereby advancing the field and providing more robust solutions for farmers. Most studies do not explore the enhanced contextual understanding and semantic richness that the word embeddings can bring to chatbot interactions, which are crucial for accurately interpreting and responding to farmers' queries. Our research aims to bridge this gap by integrating BERT, GloVe, and FastText embeddings into agricultural chatbots, thereby improving the precision and relevance of the responses and offering a more intelligent and effective assistance to farmers.

3. DATASET DESCRIPTION

Table 1. Agricultural Data

N	P	K	Temperature	Humidity	pH	Rainfall	Label
90	42	43	20.8797	82.0027	6.5030	202.9355	rice
71	54	16	22.6136	63.6907	5.7499	87.7595	maize
52	68	78	17.4850	16.9607	6.8966	86.0508	chickpea
2	61	20	22.1397	23.0225	5.9556	76.6413	kidneybeans
33	58	24	35.4579	68.7581	5.2695	108.6333	pigeonpeas

In order to create a crop recommendation system and an agriculture chatbot for this project, we used two different datasets. The first dataset was obtained from Kaggle and is a CSV file that has columns for temperature, humidity, pH, rainfall, nitrogen (N), phosphorus (P), and potassium (K), as well as a label that indicates the recommended crop. This dataset is useful for creating customized crop recommendations based on different soil and climate parameters.Table 1 shows five instances of the data. The second dataset is a JSON file with frequently asked questions about agriculture that were gathered from an agricultural website and augmented with ChatGPT to offer a broader variety. An example entry has the inquiry ”Greetings!” and the suitable response ”Greetings! ”What can I do to help you now?” The chatbot uses this dataset as its basis, which allows it to provide precise answers to commonly requested queries about agriculture.

4. METHODOLOGY

The proposed system shown in Fig 1 involves three key components:

- Crop Advisor using a stacked ensemble model
- Chatbot for agriculture queries
- End-to-End Secure Communication in Chatbot Using RSA

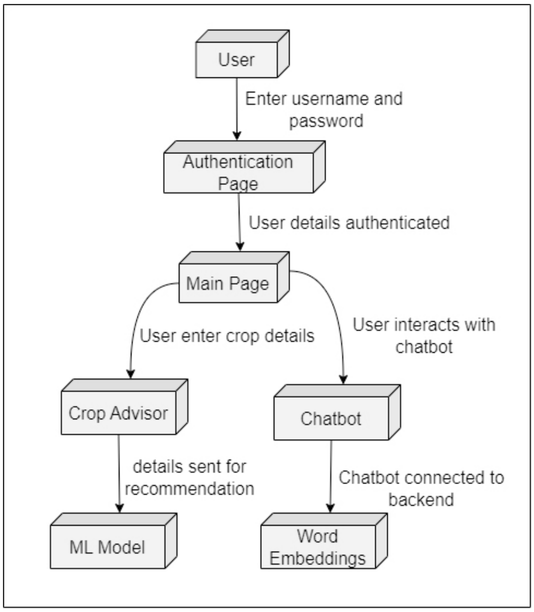

Fig. 1. Workflow of proposed Model

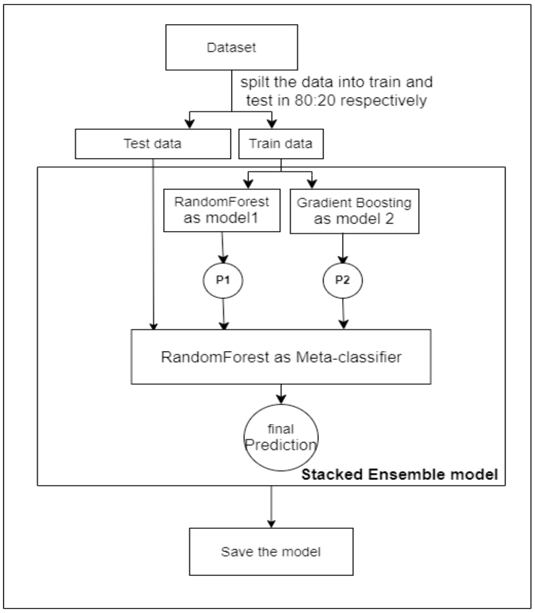

Fig. 2. Workflow of Crop Recommendation Using Stacked Ensemble Model

4.1. *Crop Advisor using a Stacked Ensemble Model*

Stacked Ensemble Model

Stacked Ensemble is a machine learning technique that combines multiple models to enhance predictive accuracy [23]. It involves training diverse base models on the same data and then using their predictions as input for a meta-model [24]. The meta-model learns to combine these predictions effectively, leveraging the strengths of each base model [25]. Stacked ensembles are often used when individual models have complementary strengths or when aiming for improved generalization on complex datasets.

Fig 2 illustrates the proposed methodology for the crop recommendation system using a stacked ensemble model . The dataset includes soil conditions (N, P, K, pH values) and weather data (temperature, rainfall, humidity), along with the corresponding suitable crops. This dataset is divided 80:20 into training and testing data, accordingly. Test data is used to assess the model's performance after it has been built, whereas training data is used to create the model.

This work utilizes two base models in a stacked ensemble model: Random Forest as Model 1 and Gradient Boosting as Model 2, with Random Forest serving as the meta-model. Algorithm 1 demonstrates the procedure of the proposed stacked ensemble model.

Algorithm 1 Stacked Ensemble Model for Crop Recommendation

```
Load and preprocess the dataset:
Load the dataset
Split the dataset into features {N, P, K, pH, temperature, rainfall, humidity} and target (crop)
Split the data into training and testing sets (80% train, 20% test)

Layer 1: Train Base Models
Model1 ← Train RandomForest on training data
Model2 ← Train GradientBoosting on training data

Layer 2: Generate Predictions on Training Data
Predictions1 ← Model1.predict(training data)
Predictions2 ← Model2.predict(training data)

Layer 3: Create Meta-classifier Features
StackedFeatures ← Stack(Predictions1, Predictions2)

Layer 4: Train Meta-classifier
MetaClassifier ← Train RandomForest on StackedFeatures

Layer 5: Make Final Predictions on Test Data
TestPredictions1 ← Model1.predict(test data)
TestPredictions2 ← Model2.predict(test data)
StackedTestFeatures ← Stack(TestPredictions1, TestPredictions2)
FinalPredictions ← MetaClassifier.predict(StackedTestFeatures)

Layer 6: Evaluate the Model
Accuracy ← Evaluate(FinalPredictions, test target)
Print("Accuracy of the stacked ensemble model: ", Accuracy)

Layer 7: Save the Trained Model
Save(MetaClassifier, "path_to_save_model")
```

Base Models

- **Random Forest Classifier (Model 1):** Trained on the training data to predict suitable crops based on provided soil and weather conditions. The Random Forest model uses multiple decision trees to provide a robust prediction.

$$\text{Prediction } RF = \frac{1}{N}\sum_{i=1}^{N}\text{Tree}_i(x)$$

where N is the number of decision trees, and $Tree_i(x)$ is the prediction from the i-th tree for input x.

- **Gradient Boosting Classifier (Model 2):** Also trained on the same training data, this model iteratively improves weak learners to enhance prediction accuracy.

$$\text{Prediction GB} = \sum_{i=1}^{M}\alpha_i h_i(x)$$

where M is the number of boosting rounds, α_i is the weight of the i-th model, and $h_i(x)$ is the weak learner for input x.

Meta-Classifier

- **Random Forest Classifier:** This classifier is trained using the predictions from the base models (Model 1 and Model 2) as input features. It combines these predictions to make a final prediction.

$$\text{Meta-Prediction} = \sum_{j=1}^{K}\text{Meta-Tree}_j(\text{Prediction RF}, \text{Prediction GB})$$

where K is the number of trees in the meta-classifier, and Meta-Tree$_j$ is the prediction from the j-th tree using predictions from Model 1 and Model 2 as inputs.

Finally, using 10-fold cross-validation (CV) on the test data, the stacked ensemble model is evaluated. and the trained model is stored for further use. This allows the system to quickly provide crop recommendations without the need for retraining. By utilizing the advantages of ensemble learning, this methodology effectively captures complex patterns in the data, leading to more accurate and reliable crop recommendations.

4.2. Chatbot for Agriculture Queries

Fig 3 presents the workflow of text preprocessing and embeddings generation for Agriculture chatbot. The JSON dataset containing agriculture-related information undergoes preprocessing steps, including tokenization, stop word removal, and lemmatization. Tokenization segregates the text into tokens, stop word removal eliminates words that do not add much meaning to the text, and lemmatization brings words down to their base form. Once preprocessed, the data is converted into numerical representations using different word embedding techniques, such as BERT, GloVe, and FastText. The generated word embeddings are stored in a Vector Database, enabling efficient similarity calculations for user queries.

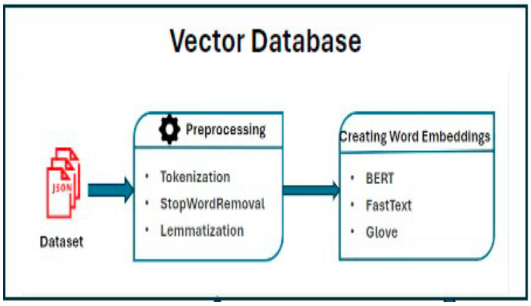

Fig. 3. Workflow Of Vector Database Creation for Agriculture Chatbot

4.3. End-to-End Secure Communication in Chatbot Using RSA

Fig 4 shows the workflow of chatbot query encryption and decryption using RSA algorithm to ensure the safe processing of user queries. When user enters a query through the chatbot interface, the query is encrypted using the server's public key. This ensures that the information exchanged is protected and cannot be easily accessed by unauthorized parties. Upon receiving the encrypted query, the server decrypts it using its private key. The decrypted query is then processed and forwarded to the Vector Database for similarity calculation. Cosine similarity is calculated

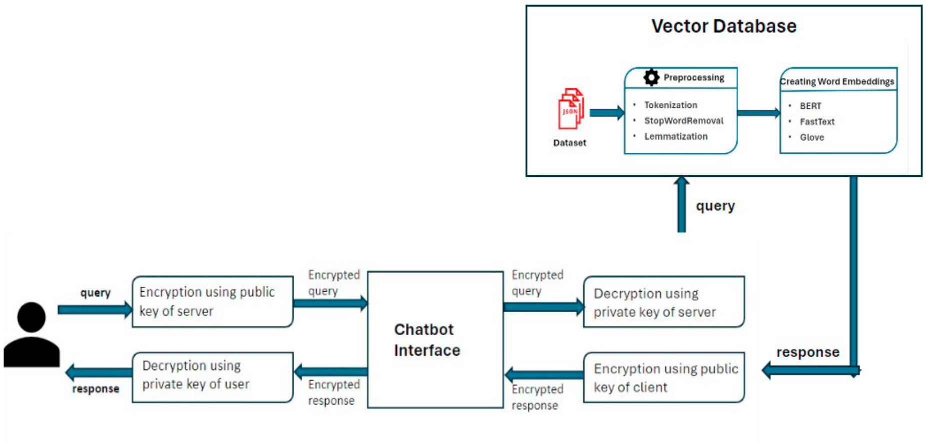

Fig. 4. Process Flow of Chatbot Query Encryption and Decryption Using RSA

between the stored word vectors present in the Vector Database and the decrypted user query. This helps in identifying the most relevant responses to the user query. Once the response is generated, it is encrypted using the user's public key. This step ensures that only the legitimate user can decrypt and read the response. The encrypted response is sent back to the user, who decrypts it using their private key and views the response.

The state diagram for the suggested system is displayed in Fig 5, detailing the steps when a user interacts with the system services. The system begins with user authentication, where users enter their username and password on the Authentication Page. The system verifies these details to authenticate the user. Upon successful authentication, users are redirected to the index.html page (Main Page) show in Fig 6, which serves as the central hub for all functions.

On the index.html page, users can access two primary features: crop recommendation and chatbot interaction. For the crop recommendation, users input soil conditions (N, P, K, pH values) and weather data (temperature, rainfall, humidity) in the crop details section. This information is sent to the backend, where it is processed by the ML model. The ML model analyzes the data and recommends the most suitable crop based on the provided conditions. The recommended crop details are then sent back to index.html for user reference.

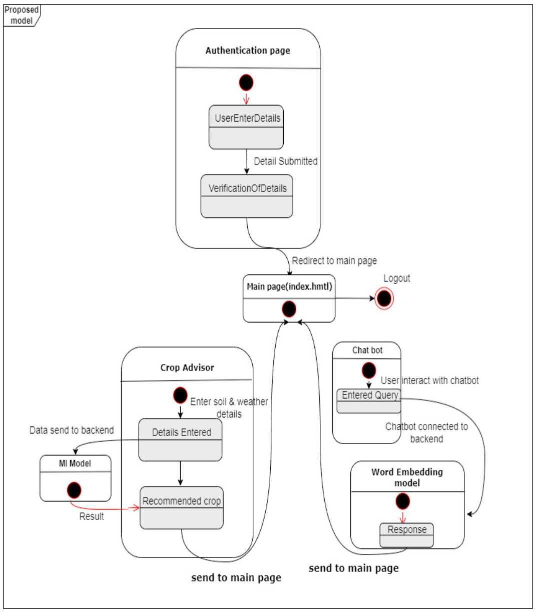

Fig. 5. State Transition Diagram of the System

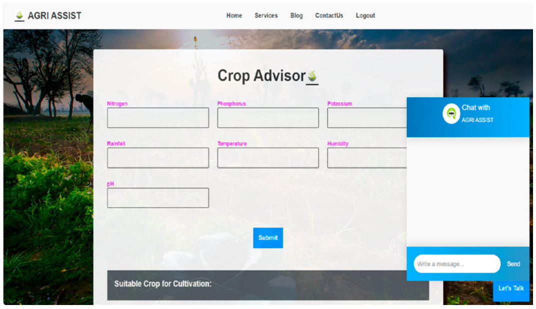

Fig. 6. Web page for user interface

In the chatbot interaction section, users can interact with the chatbot for any agriculture-related queries. The chatbot, built using word embedding techniques like BERT, GloVe, and FastText, uses the RSA encryption algorithm to secure message exchanges. When a user submits a query, the chatbot connects to the backend to fetch a response. The word embedding model processes the user query and provides relevant responses, which are then displayed on the index.html page for the user.

5. RESULTS AND ANALYSIS

Fig 7 shows the confusion matrix of proposed model. The confusion matrix for the stacked ensemble model in crop prediction demonstrates a high level of accuracy, with most predictions aligning correctly along the diagonal, indicating correct classifications. Rice is misclassified as Jute twice, and Mothbeans is misclassified once as Lentil. Every crop except Rice and Mothbeans shows perfect accuracy, with no misclassifications.

$$\text{Accuracy}(A) = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \tag{1}$$

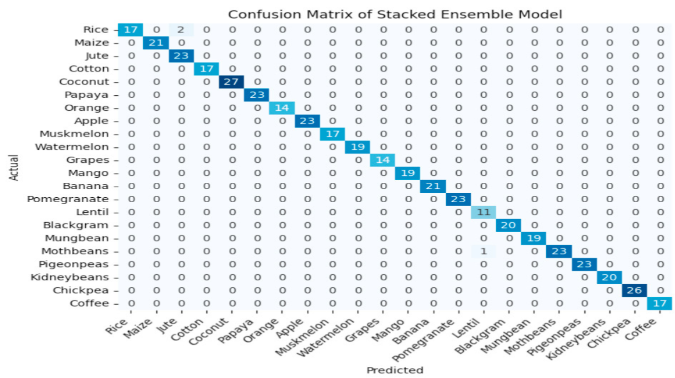

Fig. 7. Confusion matrix of stacked ensemble model

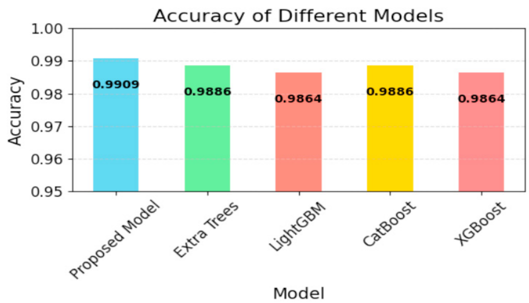

Fig. 8. Accuracies of different models

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

Test accuracy of the different machine learning models is presented in the Fig 8. The overall model ability to make correct predictions is used in estimating accuracy. The exact definition is: how many of your total predictions were correct ones. F1 Score considers both Precision and Recall at the same time. Precision and Recall help balance out each other and it's the average of them.

Table 2. Model Evaluation Metrics for Crop Prediction

Model	Accuracy	Precision	F1 Score	Recall
Proposed model	0.9932	0.9926	0.9926	0.9933
XGBoost	0.9864	0.9852	0.9861	0.9876
CatBoost	0.9886	0.9872	0.9877	0.9895
LightGBM	0.9864	0.9846	0.9852	0.9872
Extra Trees	0.9886	0.9866	0.9871	0.9895

Here we evaluate different machine learning models for crop prediction using their accuracy, precision, F1 score and recall. Table 2 presents the F1-Score and accuracies of each model. The proposed model emerged as the top performer with an impressive accuracy of 99.32%, alongside a high F1 score of 99.26%. Following closely, the XGB-Classifier, CatBoostClassifier, LightGBM, and Extra Trees also demonstrated strong predictive capabilities, achieving accuracies of 98.64%, 98.86%, 98.64%, and 98.86% respectively. These models exhibited robustness across metrics, indicating their suitability for accurate crop type predictions based on given features. Each model's performance reflects its efficacy in handling the complexities of crop prediction tasks, offering reliable insights for agricultural decision-making.

Table 3. State of the Art (SOTA) Analysis

Algorithm	**Precision**	**Recall**	**F1 Score**	**Accuracy**
SVM[9]	97.80%	97.78%	97.78%	97.78%
XGBoost[26]	-	-	-	99.31%
Ensemble[27]	99%	99%	99%	99%
Proposed Model	99.26%	99.33%	99.26%	99.32%

The Table 3 shows the comparative study of previous related works. Shah et.al[9] achieved an accuracy of 97.78% using SVM. Similarly, Hossain et.al[26], Gosai[27] performed XGBoost,Ensemble models to attain an accuracies of 99.31% and 99% respectively.

Fig. 9. Client side communication using RSA algorithm

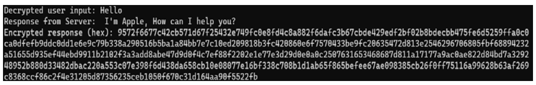

Fig. 10. Server side communication using RSA algorithm

Fig 9 and Fig 10 illustrate the detailed communication process between a client and server in an RSA-encrypted chatbot system. Using the public key of the server, the client first inputs "Hello," which is subsequently encrypted. The server receives this communication, which is encrypted. After receiving the information, the server uses its private key to decrypt the message and correctly interprets it as "Hello." "I'm Apple, How can I help you?" is the response that the server then creates and encrypts using the client's public key. The client receives this encrypted response back, which it decrypts to display the message from the server. Encryption, transmission, and decryption procedures are verified by the displays of the encrypted and decrypted messages on both ends. By using RSA encryption to protect data confidentiality, the system successfully shows safe and verified communication.

Table 4. Model Comparison

Model	**Cosine Similarity**	**Execution Time (seconds)**
BERT	0.85689	0.5638
GloVe	0.85753	0.0219
fastText	0.88629	0.0244

The cosine similarity and execution time are used as parameters to generate the results for the three models. Table 4 shows the cosine similarity and execution time for each model. The cosine similarity of the contextual model BERT, which ran in 0.5638 seconds, was 0.85689. This demonstrates how well BERT understands the context and semantics of the provided question. With a substantially shorter execution time of 0.0219 seconds and a slightly higher cosine similarity of 0.85753, the GloVe model—which is based on global word co-occurrence statistics—performed better. As it can be observed, GloVe captures semantics fairly well and is incredibly computationally efficient.

Among the models, the fastText model which extended the word2vec model by including subwords yielded the highest cosine similarity of 0.9985 with an execution time of 0.0244 seconds. This suggests that fastText can learn the intricate semantic relationship between words in a text despite its efficiency. In general, fastText has the highest cosine similarity and moderate time complexity, and GloVe is not very different from fastText in terms of accuracy and time consumption. It can therefore be concluded that BERT, though slightly lagging behind in cosine similarity and time taken, is a strong contender for constructing a meaningful text representation.

Based on the findings, fastText provides better cosine similarity and requires lesser time for the query as compared to BERT and GloVe. As pointed out above, these results suggest that fastText is particularly appropriate in scenarios where both accuracy and time are of the essence.

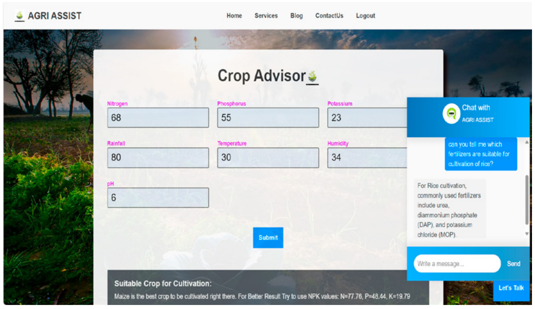

Fig. 11. Crop recommendation and Agricultural chatbot

Fig 11 shows recommendation of crop and chatbot on crop advisor website. This allows users to enter information about their crops, such as soil pH, rainfall, and temperature, and receive crop recommendations. Based upon the information given by the user, the website suggests that maize is the best crop to be cultivated in the user's location. The user has also asked for information about fertilizers used for rice cultivation. The chatbot recommends that the user to apply urea, diammonium phosphate (DAP), and potassium chloride (MOP) fertilizers.

6. CONCLUSION AND FUTURE SCOPE

The proposed AI Agriculture Assistant system successfully integrates a crop recommendation system and a chatbot for handling agricultural queries, offering a wide service package that will be beneficial to farmers as well as other agricultural enthusiasts. Utilizing the FastText model for faster query results ensures quick and accurate responses, while the stacked ensemble model for crop recommendations leverages multiple algorithms to enhance prediction accuracy. The chatbot, employing the RSA encryption algorithm, guarantees secure communication, safeguarding user data and interactions.The system is deployed using Flask, enabling seamless, real-time interaction between users and the AI assistant. This comprehensive solution is designed to provide reliable and tailored agricultural assistance, significantly improving decision-making processes for crop selection and management. By combining state-of-the-art machine learning models with robust encryption and user-friendly deployment, the AI Agriculture Assistant system stands out as a cutting-edge tool in modern agriculture.

Future work includes adding multilingual support to cater to a diverse user base and developing a mobile app for enhanced accessibility. Enhancing security features will further protect user data and ensure the integrity of the system. These improvements aim to make the AI Agriculture Assistant even more versatile and user-friendly.

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